

CITYCARE GENERAL HOSPITAL

INTENSIVE CARE UNIT STANDARD
OPERATING PROCEDURE

Critical Care Admission, Monitoring & Management

SOP Code	SOP-ICU-005
Department / Unit	Intensive Care Unit (ICU) / High Dependency Unit (HDU)
Document Version	2.1
Effective Date	01 January 2025
Review Date	31 December 2025
Prepared By	Quality Assurance & Clinical Governance Team
Approved By	Chief Medical Officer, CityHospital Group
Applicable To	All Clinical and Administrative Staff
Classification	CONFIDENTIAL — Internal Use Only

NOTE: This document is a controlled Standard Operating Procedure. Any printed copy is uncontrolled unless stamped 'CONTROLLED COPY'. For the latest version, refer to the Hospital Intranet Document Management System.

TABLE OF CONTENTS

1	Purpose & Scope
2	Definitions & ICU Level Classification
3	Roles & Responsibilities
4	ICU Admission Criteria & Workflow
5	Patient Monitoring Standards
6	Mechanical Ventilation Protocol
7	Infection Control in ICU
8	Sedation, Analgesia & Delirium Management
9	Family Communication & Visiting Policy
10	ICU Discharge & Step-Down Protocol

1. PURPOSE & SCOPE

This SOP governs the operation of the CityHospital Intensive Care Unit and High Dependency Unit, covering admission criteria, continuous monitoring standards, therapeutic protocols, and safe step-down procedures. The ICU and HDU provide the highest levels of physiological monitoring and organ support available within the hospital.

- Establish clear admission and discharge criteria to optimise critical care resource utilisation.
- Define continuous monitoring standards to detect and respond to deterioration rapidly.
- Ensure evidence-based management of mechanical ventilation, sepsis, and multi-organ dysfunction.
- Promote patient dignity, family inclusion, and psychological wellbeing in the ICU environment.
- Prevent ICU-acquired complications: ventilator-associated pneumonia, CLABSI, and pressure injuries.
- Maintain nurse-to-patient ratios appropriate to care level at all times.

2. DEFINITIONS & ICU LEVEL CLASSIFICATION

Care Level	Setting	Nurse:Patient Ratio	Description
Level 1	Ward	1:4 to 1:6	At risk of deterioration; enhanced monitoring, not organ support
Level 2	HDU	1:2	Single-organ support (e.g., vasopressors or CPAP alone)
Level 3	ICU	1:1	Multi-organ support including mechanical ventilation

- **APACHE II:** Acute Physiology and Chronic Health Evaluation — predicts ICU mortality risk.
- **SOFA:** Sequential Organ Failure Assessment — tracks multi-organ dysfunction over time.
- **RASS:** Richmond Agitation-Sedation Scale — sedation depth assessment tool.
- **CAM-ICU:** Confusion Assessment Method for ICU — validated delirium screening tool.
- **VAP:** Ventilator-Associated Pneumonia — most common ICU-acquired infection.
- **CLABSI:** Central Line-Associated Bloodstream Infection.
- **HFNO:** High-Flow Nasal Oxygen — non-invasive respiratory support.
- **NIV:** Non-Invasive Ventilation (e.g., BiPAP, CPAP).
- **RSI:** Rapid Sequence Induction — emergency intubation technique.

3. ROLES & RESPONSIBILITIES

Role	Department	Responsibilities
ICU Consultant/Intensivist	ICU	Overall clinical leadership, complex decisions, family meetings, discharge planning
ICU Registrar	ICU	Daily ward rounds, clinical orders, procedures, out-of-hours management
ICU Charge Nurse	ICU	Shift coordination, staff allocation, safety checks, escalation
Bedside ICU Nurse	ICU	Continuous monitoring, drug titration, care bundle compliance, family support
Physiotherapist	Allied Health	Ventilator weaning, early mobilisation, chest physiotherapy
Clinical Pharmacist	Pharmacy	Drug review, renal/hepatic dose adjustment, TPN preparation, reconciliation
Dietitian	Nutrition	Enteral or parenteral nutrition plan, caloric targets, monitoring
Infection Control Nurse	IPC	CLABSI and VAP bundle audits, environmental surveillance, outbreak response

4. ICU ADMISSION CRITERIA & WORKFLOW

4.1 Admission Criteria — Level 3 ICU

- Actual or anticipated need for invasive mechanical ventilation.
- Haemodynamic instability: MAP below 65 mmHg despite adequate fluid resuscitation, requiring vasopressor support.
- Acute respiratory failure: SpO₂ below 90% on FiO₂ above 0.6, or rising RR above 30/min with respiratory distress.
- Acute neurological deterioration: GCS of 8 or below, or acute stroke with airway compromise.
- Post-operative monitoring: high-risk surgery patients per anaesthetic or surgical decision.
- Septic shock, severe DKA, or acute liver failure with multi-organ system involvement.
- Severe electrolyte or metabolic disturbance requiring continuous monitoring and IV replacement therapy.

4.2 ICU Admission Workflow

Step 1. Referring physician contacts ICU consultant; SBAR-based clinical handover completed verbally.

Step 2. ICU bed confirmed by charge nurse; receiving bay prepared with appropriate monitoring and equipment.

Step 3. Patient transferred with uninterrupted monitoring; infusions and ventilator settings maintained during transport.

Step 4. Receiving ICU nurse performs full systems assessment on arrival; APACHE II and SOFA scores calculated.

Step 5. ICU admission orders entered in EMR; goals of care discussed and documented.

Step 6. Family notified of ICU admission within 2 hours; family liaison nurse assigned.

NOTE: ICU admission rejected by the intensivist must be documented in the EMR with clear clinical rationale. The referring physician retains responsibility until formal acceptance.

5. PATIENT MONITORING STANDARDS

All Level 3 ICU patients require continuous monitoring of the following parameters. Deviations from target ranges must be escalated to the ICU physician immediately.

Parameter	Target Range	Monitoring Frequency
Arterial Blood Pressure	MAP 65–100 mmHg	Continuous invasive ABP or 15-min NIBP
Heart Rate	60–100 bpm	Continuous ECG monitoring
SpO2	94% or above (88–92% in COPD)	Continuous pulse oximetry
Urine Output	0.5 ml/kg/hr or above	Hourly via urinary catheter
Core Temperature	36.0–38.0 degrees C	Every 2 hours
Blood Glucose	6–10 mmol/L	Every 1–4 hours (insulin infusion protocol)
ABG / Ventilator Parameters	pH 7.35–7.45	Every 4 hours or after any change
RASS Score	-2 to 0 (light sedation target)	Every 2 hours
CAM-ICU Delirium Screen	Negative	Every 8 hours (once per shift)
Central Venous Pressure (if applicable)	5–12 cmH2O	Continuous if CVP line in situ

6. MECHANICAL VENTILATION PROTOCOL

6.1 Initiation

- Indication for intubation confirmed by ICU consultant; RSI performed by senior physician.
- Lung-protective ventilation settings applied: tidal volume 6 ml/kg IBW, PEEP 5–8 cmH₂O, FiO₂ titrated to SpO₂ target.
- Chest X-ray confirmed within 30 minutes of intubation; endotracheal tube position formally documented.
- VAP prevention bundle activated immediately: HOB elevation to 30–45 degrees, oral care every 4 hours, subglottic suction, and daily sedation vacation.

6.2 Ventilator Alarms

- High pressure alarm: respond within 60 seconds; suction, check for cough, kinking, or pneumothorax.
- Low pressure or disconnection alarm: reconnect circuit immediately; confirm lung ventilation bilaterally.
- Apnoea alarm: manual ventilation commenced; physician notified immediately.
- All alarm responses documented in EMR with time, action taken, and patient response.

6.3 Ventilator Weaning Protocol

Step 1. Daily weaning assessment: patient responsive (RASS -1 or above), haemodynamically stable, FiO₂ of 0.5 or below, PEEP 8 or below.

Step 2. Daily spontaneous awakening trial (SAT): sedation held each morning to assess neurological status.

Step 3. Spontaneous breathing trial (SBT) performed for 30 to 120 minutes on minimal support (PS 5–8 cmH₂O).

Step 4. Pass criteria: RR below 35, SpO₂ above 90%, no respiratory distress, no haemodynamic deterioration.

Step 5. Failed SBT: return to pre-SBT settings immediately; document reason; reassess following day.

Step 6. Successful SBT: extubation planned and performed within 2 hours; post-extubation monitoring for 2 hours.

7. INFECTION CONTROL IN ICU

7.1 CLABSI Prevention Bundle

- Full sterile barrier precautions for all central venous line insertions (sterile gown, gloves, mask, cap, and full body drape).
- Chlorhexidine-alcohol skin antisepsis applied; 2-minute dry time observed before needle insertion.
- Daily line necessity review: clinical indication documented each day; line removed when no longer required.
- Dressing change every 7 days or when soiled; chlorhexidine-impregnated dressing used as standard.
- Daily chlorhexidine bathing for all ICU patients to reduce skin microbial colonisation.
- Hub decontamination protocol: 15-second scrub before every access; positive pressure caps applied.

7.2 VAP Prevention Bundle

- Head-of-bed elevation maintained at 30 to 45 degrees at all times in intubated patients.
- Oral care with chlorhexidine solution performed every 4 hours; documented on care bundle chart.
- Subglottic secretion drainage: continuous or intermittent drainage tube used for all intubations expected to exceed 48 hours.
- Ventilator circuit changes only when visibly soiled or malfunctioning; not on a routine schedule.
- Hand hygiene compliance audited weekly; target greater than 95% compliance.

7.3 Pressure Injury Prevention

- Braden scale assessed on ICU admission; all intubated patients placed on pressure-redistributing mattress.
- 2-hourly repositioning chart completed and signed by nurse; heel offloading applied to all intubated patients.
- Nutrition optimised: protein target 1.2–2g/kg/day; dietitian review within 24 hours of admission.

8. SEDATION, ANALGESIA & DELIRIUM MANAGEMENT

8.1 Analgesia First — eCASH Principle

- Pain assessed and managed before sedation is considered (Early Comfort using Analgesia, minimal Sedatives and maximal Humane care).
- Target RASS of -1 to 0 for all mechanically ventilated patients unless specific clinical indication for deeper sedation.
- Opioid infusions titrated to NRS pain score; patient-reported pain assessed at each nursing assessment.
- Non-opioid analgesia (paracetamol, ketamine, regional blocks) used to minimise opioid requirements.

8.2 Sedation Management

- Daily spontaneous awakening trial (SAT): sedation held each morning; patient reassessed; sedation restarted if distressed.
- Sedation type and dose reviewed daily at ICU ward round; step-down to lowest effective dose.
- Propofol infusion syndrome monitoring: triglycerides and CK measured every 48 hours in patients on high-dose propofol.

8.3 Delirium Prevention & Management

- CAM-ICU screening every 8 hours; all positive results escalated to ICU physician.
- Non-pharmacological delirium prevention bundle: day-night light cycle, noise reduction at night, early mobilisation, family presence, reorientation.
- Minimise deliriogenic medications: anticholinergics, benzodiazepines, and unnecessary polypharmacy reviewed daily.
- Haloperidol or quetiapine prescribed only per ICU consultant order for hyperactive delirium causing safety risk.

9. FAMILY COMMUNICATION & VISITING POLICY

- ICU consultant or registrar provides family clinical update within 12 hours of admission and at least daily thereafter.
- MDT family meeting conducted within 48 hours for patients expected to remain in ICU for more than 72 hours.
- Standard visiting hours: 10:00–12:00 and 17:00–20:00; maximum 2 visitors at bedside simultaneously.
- Open visiting for immediate next-of-kin when patient condition is immediately life-threatening.
- Family liaison nurse appointed for all long-stay ICU patients (stay exceeding 5 days).
- Written ICU family guide provided in patient's language within 24 hours of admission.
- Goals of care discussion documented in EMR; advance directives reviewed and honoured.
- Psychological support offered to family members of critically ill patients; chaplaincy and social work involved as needed.

10. ICU DISCHARGE & STEP-DOWN PROTOCOL

10.1 Step-Down Criteria — ICU to HDU or Ward

- Haemodynamically stable without vasopressor support for a minimum of 24 hours.
- Extubated and maintaining SpO₂ above 94% on 4 L/min or less supplemental oxygen for at least 4 hours.
- Delirium resolved or manageable without continuous sedation; patient communicating.
- Nutritional needs met by enteral or oral route; parenteral nutrition discontinued.
- Accepting physician on receiving ward or HDU identified; formal handover planned.
- No anticipated need to return to ICU-level care within 24 hours as assessed by intensivist.

10.2 Step-Down Handover Workflow

Step 1. ICU charge nurse contacts receiving ward or HDU; confirms bed availability and nursing competency.

Step 2. Full SBAR handover: active IV lines, infusions, monitoring requirements, and all active medications communicated.

Step 3. ICU discharge summary completed by intensivist in EMR; printed copy accompanies patient.

Step 4. Patient and family informed of transfer plan; reassurance provided and questions answered.

Step 5. ICU follow-up outpatient clinic appointment offered for all patients ventilated for more than 48 hours.

NOTE: ICU step-down should be planned during business hours (08:00–20:00) where clinically possible to ensure adequate senior nursing cover on the receiving ward at time of handover.